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Optimisation of Traffic Lights for Signalised Traffic Flow on Different Road Geometries

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Abstract

This report studies the optimisation of through-flow on signalised traffic networks using a discrete cellular automaton (CA) model. A Nagel-Schreckenberg (NaSch) single-lane model is developed and then extended to a two-lane road with the asymmetric Chowdhury-Wolf-Schreckenberg (CWS) lane-change rule with heterogeneous slow/fast vehicles. Afterwards we extend the model in another way with a two-dimensional city grid with a Brockfeld diagonal-offset green wave and turning junctions. Through-flow is measured under periodic and open boundary conditions, averaged across multiple random seeds, and a sensitivity analysis is performed on the grid-with-turning model.

The simulations show that the green wave strategy consistently outperforms the other two strategies, especially as we increase the number of lights or grid size. We also see that green wave survives increased lane changing and turning probabilities, which suggests that it is a strategy better for optimising real-life traffic flow. The sensitivity analysis also suggests that the most important parameters to control are the ones related to the human behaviour such as the dawdling probability and turning probability, which may be harder to control in real-life but may be the ones causing the most congestion. We also see that increasing the speed limit can increase flow up to a certain point, after which it plateaus and slightly decreases, which suggests that there is an optimal speed limit for traffic flow under the green wave strategy.

Key Words: Traffic flow, Cellular automaton, Nagel-Schreckenberg, Green wave, CWS lane-change, Sensitivity analysis

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1 Introduction

1.1 Real-world Context

Urban Traffic Congestion affects developed and developing countries alike. It arises from a supply-demand imbalance in road capacity and drives fuel consumption, greenhouse emissions, poor air quality and accidents [1]

In the UK, this issue is particularly prevalent. In the 2019, UK drivers lost an average of 115 hours each to road traffic congestion [2].

Since traffic lights are already widely used, optimising their timing and coordination is a low-cost way to improve traffic flow and reduce congestion.

1.2 Why discrete cellular automata

We can leverage the use of cellular automata to model traffic flow, this is because they are a powerful tool for simulating complex systems and can capture the dynamics of traffic flow in a simple and efficient way. Cellular Automata also allow us to assess the impact of different traffic light strategies on traffic flow, and can help us to identify the most effective strategies for reducing congestion.

Traffic cellular automata models have been widely used in traffic flow research, and have been shown to be effective in capturing the dynamics of traffic flow and congestion. Examples of cellular automata models include the Nagel-Schreckenberg (NaSch) model and the Biham-Middleton-Levine (BML) model [3, 4].

For this study, we will use the NaSch model due to its simplicity and more clearly explaining the effects on traffic lights in the system. The NaSch model supports the three extensions we study which include multiple traffic lights, a second lane and a two-dimensional city grid.

1.3 Aims and Research Questions

In this report, we will explore three different contexts of traffic flow, exploring how different traffic light layouts affect the system. The first context will be a single lane system, where we vary traffic light number and traffic light strategies [3]. The second one will be a two lane system, where we will explore the effect of varying traffic light number and traffic light strategies with the introduction of different vehicle types on the system [5, 6]. The last one is a $N \times N$ grid extension of the NaSch model, where we will explore the effect of varying traffic light strategies on the system, as well as increasing the road size [7].

The unifying question across these three is light configuration and strategy, we want to understand how different traffic light configurations and strategies can affect traffic flow and congestion across all 3 contexts. The three geometries are shown in Figure 1.

We will address four research questions:

1. How does single-lane through-flow scale with the number of traffic lights, the cycle length, the green fraction and the inter-light phase offset?
2. When does the green-wave strategy outperform synchronised lights and why?
3. Does asymmetric lane changing complement the green-wave or fight it?
4. Does the green-wave coordination strategy stay the best strategy when extending from 1-D to a 2-D city grid?

2 Background

2.1 Continuum vs Discrete Traffic Flow

Analytical traffic models can be either continuum or discrete. Early examples of traffic flow models were based on continuum mechanics. Continuum models treat traffic as a continuous fluid, where instead of modelling individual vehicles, the vehicle density, mean velocity and flow are all modelled as continuous functions of space and time. Lighthill and Whitham represented the road as a one-dimensional fluid with flow q (vehicles per hour), concentration k (vehicles per mile) and posited as the fundamental hypothesis of their theory that the flow q is a function of k alone [8]. The continuum models are useful for understanding traffic flow at a macroscopic scale, formation of traffic jams and the effects of traffic lights but cannot capture the microscopic behaviour of individual vehicles and the interactions between them. Discrete models on the other hand address this weakness by treating everything as a finite integer such as space, time, velocity and the vehicles themselves, as we discuss next.

2.2 The NaSch cellular automaton

Nagel and Schreckenberg proposed a stochastic discrete cellular automaton where a road is represented as a one-dimensional lattice of cells, each of which can be occupied by a single vehicle or be empty. Each vehicle carries an integer velocity v in the range $\{0, 1, \dots, v_{\max}\}$ and the system is updated in discrete time steps according to a set of rules that govern the movement of the vehicles. The rules are as follows:

1. **Acceleration:** If the velocity of a vehicle is less than the maximum velocity $v_{\max}$, and if the distance to the next car is larger than $v + 1$, the velocity is increased by 1, $[v \rightarrow v + 1]$.
2. **Deceleration:** If a vehicle at cell i sees the next vehicle at cell $i + j$ (with $j \leq v$), it reduces its speed to $j - 1$, i.e. $[v \rightarrow j - 1]$.
3. **Randomisation:** With probability p , the velocity of each vehicle (if greater than zero) is decreased by one $[v \rightarrow v - 1]$.
4. **Movement:** Each vehicle is advanced v cells.

Here p is the probability of random deceleration, which is a key parameter that controls the behaviour of the system. Through the four rules above, the general properties of single lane traffic are modelled on the basis of integer valued probabilistic cellular automaton rules. This simple model is effective in capturing the nontrivial and realistic behaviour of traffic flow. Step 3 is particularly important for simulating realistic traffic flow, without it the dynamics is purely deterministic, and from any initial configuration the system relaxes to a stationary pattern shifted backwards one cell per timestep [3, 9]. The model also accounts for natural velocity fluctuations due to human behaviour or due to varying external conditions. The four steps above are unchanged across the single lane and two lane scenarios.

2.3 Two-Lane Extensions

Rickert et al. proposed a two-lane extension to the NaSch model, where vehicles can change lanes according to a set of rules that depend on the local traffic conditions. The lane-change sub-step runs before the NaSch update, with both sub-steps applied as a strict parallel update against the configuration at the start of each time step [5].

Chowdhury et al. extended Rickert's work to include two different types of vehicles, slow (e.g. a truck) and fast (e.g. a car), and proposed two variants of the lane change rule: a *symmetric* variant in which the same rules apply for both lanes, and an *asymmetric* variant modelled after the German Autobahn convention where the right (slow) lane is preferred and the left (fast) lane is used for overtaking only [6]. In this report we will explore the asymmetric variant of the lane change rule.

For the symmetric version, the rules for the parallel updating of the speeds and positions of the vehicles are as follows: first, each vehicle is considered for the possibility of a lane changing and if the rules allow the lane change, the changes are implemented; second the speed and position of each vehicle is updated following the rules for the forward movement in the current lane. A vehicle changes lane with probability p_{lc} provided:

1. the gap in front in the current lane is less than $v + 1$

2. the gap in front in the other lane is larger than the gap available in front in the current lane
3. the nearest-neighbour site in the other lane is empty
4. the maximum possible speed of the vehicle behind in the other lane is smaller than the gap in between

In the asymmetric version, the change from the slow lane to the fast lane uses the four conditions of the symmetric version, but the change from the fast lane back to the slow lane is weaker: a vehicle returns to the slow lane as long as conditions (3) and (4) above are satisfied and the slow-lane gap is sufficient, without requiring (1) and (2) to be satisfied.

The rules for updating the speeds and positions of the vehicles in the current lane are identical to those in the NaSch single lane model in Section 2.2 [6]. Lane-changing alters the design space for light optimisation, since cars can route around a queue at one light by switching lane.

In this report we will explore the asymmetric version due to it providing a more realistic model of lane changing behaviour in the UK.

2.4 Two-Dimensional Grid Extension

Biham, Middleton and Levine (BML) introduced a simple two-dimensional cellular automaton model for city traffic, where horizontal and vertical moving cars share a lattice geometry with one car per cell. This allowed for stable states where there would be congestion in the model[4, 10]. The model captured the jamming in grid traffic but ignored velocity and dawdling.

Brockfeld et al. combined BML's grid topology with the per lane NaSch dynamics to create a more realistic model of city traffic, where each cell can be occupied by a vehicle moving in one of two directions, either horizontally or vertically, and the traffic lights at the intersections control the flow of vehicles through the lattice [7].

2.5 Light Strategies

Traffic lights are a common feature of urban traffic networks, and their timing can have a significant impact on traffic flow and congestion.

Three coordination strategies are explored in this report: synchronised lights (all switch in phase), the green wave (offsets are set so that a free flow vehicle meets each light when it turns green) and random offsets (each light's offset parameter is chosen from a uniform distribution). Brockfeld et al. show that the green wave gives much improvement to the flow of traffic in comparison to the synchronised strategy. Random offset outperforms the synchronised strategy at low densities for small cycle times, and at high densities for all cycle times [7].

3 Model Description and Boundary Form

3.1 Model Assumptions

The following assumptions have been made:

- The system has a discrete space and time
- Single occupancy of cells
- Integer velocities with global ceiling
- No overtaking in the single lane system
- In the single-lane model all drivers are identical: every car follows the same four NaSch rules with the same dawdling probability p .
- In the two-lane model all drivers obey the same lane-change rule with a global lane-change probability p_{lc} .
- No anticipation of traffic conditions, drivers only react to the current state of the system and do not anticipate future states.
- No reverse movement, on the single and two lane roads vehicles can only move forward and cannot reverse or change direction. On the two-dimensional grid, vehicles can only move forward, unless they are at an intersection, where they can turn onto the perpendicular road with probability p_{turn} .
- At the start of each timestep the road state is captured as a fixed snapshot; every vehicle then computes its update (rules 1-4) from that snapshot alone, and the resulting positions and velocities are written to the lattice at the end of the step. The order in which cars are processed therefore does not affect the outcome.
- Heterogeneous traffic is only admitted in the two lane model, so the single lane and two-dimensional grid models are homogeneous with a single velocity ceiling v_{max} .
- In the 2D grid model all drivers obey the same NaSch rules with the same dawdling probability p , and same turning probability p_{turn} .

These assumptions produce a tractable cellular automata model for traffic flow, while still capturing key features of real traffic such as congestion and the effects of traffic lights.

3.2 Road Geometries and Light Configuration

Three road geometries are studied in this report: a single lane road with N traffic lights, a two lane road with mixed vehicle types (cars and trucks) with N traffic lights, and a two-dimensional city grid where we have with $N \times N$ intersections with N^2 traffic lights.

Different light strategies are studied in these cases where we vary timing, spacing and synchronisation of the lights. To align with UK driving conditions, for the asymmetric two lane case, we assume that the left lane is the slow lane and the right lane is the fast lane, used for overtaking as the left lane is generally more preferred by drivers in the UK, and the right lane is used for overtaking and faster traffic, this is the opposite of the German system where the left lane is the fast lane and the right lane is the slow lane, this is done to make the results more relevant to UK traffic conditions.

The three road geometries studied in this report are shown in Figure 1.

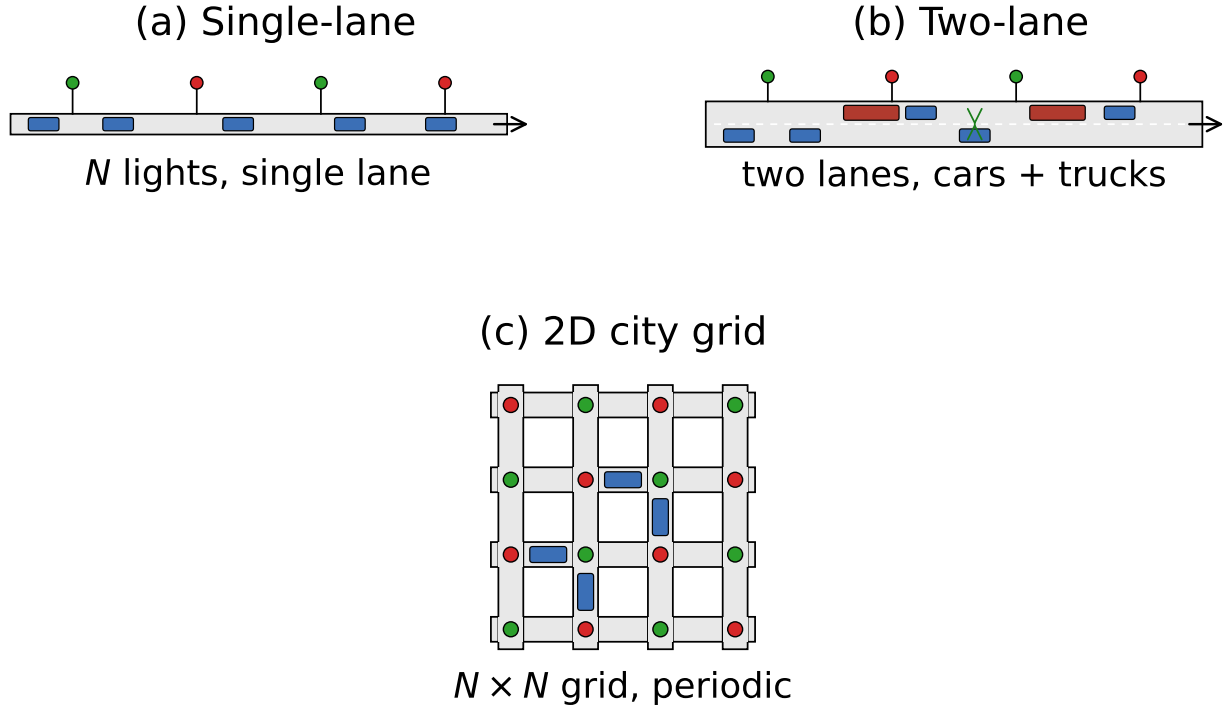


Figure 1: The three road geometries studied in this report. (a) Single-lane road with N traffic lights, cars travel left-to-right. (b) Two-lane road with mixed vehicle types (cars in blue, trucks in red) and N lights, the green arrow marks a lane change. (c) Two-dimensional city grid with N^2 traffic lights, one at each intersection. Cell counts, light spacing, and cycle parameters are set in the corresponding model section. Schematics are illustrative

Figure 1 summarises the three geometries: in the single lane case (a), cars travel from left to right past N lights; in the two-lane case (b) cars travel from left to right in both lanes, with the slow lane (the higher lane in the schematic) being the main lane and the lower lane being the fast lane used for overtaking, similarly there are N traffic lights.

The two-dimensional grid case (c) extends the model to two dimensions: N west to east lanes intersect with N south to north lanes, forming a grid of N^2 intersections, each with a traffic light. Cars can turn onto the perpendicular road at an intersection with probability p_{turn} .

3.3 Single Lane Lattice and State

The single lane road is represented as a one-dimensional lattice of length L , indexed from $i = 0$ at the entry to $i = L - 1$ at the exit, with vehicles travelling from left to right. Each cell can be either empty or occupied by a vehicle which carries an integer velocity $v \in \{0, 1, \dots, v_{\text{max}}\}$ in cells per timestep. Following the encoding used throughout the implementation, an empty cell is represented by -1 and an occupied cell is represented by the velocity of the vehicle occupying it, so the state of the road at time t is the integer array

$$\text{road}(t) \in \{-1, 0, 1, \dots, v_{\text{max}}\}^L. \quad (1)$$

The gap g_i ahead of a vehicle at cell i is the number of empty cells between it and the next obstacle ahead, where an obstacle is either the next vehicle or a red traffic light.

The four NaSch rules are applied as a parallel update over the snapshot at the start of each timestep, advance the state by one step. These rules are described in the next section.

3.4 NaSch Update Rules

Each timestep applies the four NaSch rules of Section 2.2 as a strict parallel update: for a vehicle at cell i with velocity v and gap g_i to the next obstacle ahead, the rules is as follows:

$$v \leftarrow \min(v + 1, v_{\max}), \quad (2)$$

$$v \leftarrow \min(v, g_i), \quad (3)$$

$$v \leftarrow \max(v - 1, 0) \quad \text{with probability } p, \quad (4)$$

$$i \leftarrow i + v. \quad (5)$$

Equation 4 is the stochastic element of the cellular automaton, without it the dynamics are deterministic and relax to a steady state pattern with no traffic jams. The rule creates the velocity fluctuations responsible for the spontaneous jam formation observed in real traffic flow, and is key for simulating realistic traffic flow. [3, 9].

3.5 Model Scaling and Interpretation

The model can be scaled so that each car occupies around 7.5 m of road which is the length of one cell. The mean free flow velocity is $v_{\text{free}} = v_{\max}p \approx 4.5$ cells per time step, which corresponds to around 120 km/h (75 mph) in real world terms, this is a reasonable speed for free traffic on a motorway. Combining the cell length and the free flow velocity gives a real time per time step

$$7.5 \frac{\text{m}}{\text{cell}} \times 4.5 \frac{\text{cells}}{\text{time-step}} / (120/3.6) \frac{\text{s}}{\text{m}} \approx 1.0 \text{ s} \quad (6)$$

[3]. As this model is scalable to different scenarios there is not a need to non-dimensionalise the model, we can simply use the above scaling to interpret the results in real world terms

3.6 Two-Lane Extension and The CWS Lane-Change Rule

The two-lane road extends the single lane state of Section 3.3 to a $(2, L)$ integer array using the same $\{-1, 0, \dots, v_{\max}\}$ encoding per cell. Lanes are indexed so that lane 1 is the slow left hand lane and lane 0 is the fast right hand lane used for overtaking, following the UK driving convention mentioned in 3.2.

Each timestep applies the Chowdhury-Wolf-Schreckenberg (CWS) lane change rule step, afterwards the NaSch step. Each of these steps follows the parallel-update scheme described earlier on.

For the symmetric variant, a vehicle at cell i in lane l considers changing to the other lane $l' = 1 - l$ with probability p_{lc} , and if the following four conditions are satisfied:

$$g_{\text{own}}^{\text{ahead}} < \min(v + 1, v_{\max}^{(i)}), \quad (\text{C1: tailing car in front})$$

$$g_{\text{other}}^{\text{ahead}} > g_{\text{own}}^{\text{ahead}}, \quad (\text{C2: improvement})$$

$$\text{cell}_{\text{other}}(i) = -1, \quad (\text{C3: target empty})$$

$$g_{\text{other}}^{\text{behind}} > v_{\max}, \quad (\text{C4: safe rear})$$

where $g_{\text{own}}^{\text{ahead}}$ and $g_{\text{other}}^{\text{ahead}}$ are the gaps ahead in the current and other lane respectively, $g_{\text{other}}^{\text{behind}}$ is the gap to the nearest car behind in the other lane, and $v_{\max}^{(i)}$ is the per-vehicle velocity ceiling under heterogeneous traffic (equal to $v_{\max}$ for homogeneous traffic). The safe rear condition uses the global velocity ceiling $v_{\max}$ as the velocity of the trailing vehicle is not known to the focal vehicle.

For the asymmetric variant, the change from the slow (left) lane to the fast (right) lane uses the same four conditions as the symmetric variant. The change from the fast (right) lane back to the slow (left) lane is weaker: a vehicle returns to the slow lane with probability p_{lc} as long as:

$$g_{\text{other}}^{\text{ahead}} \geq v + 1, \quad (\text{C2': slow-lane room})$$

$$\text{cell}_{\text{other}}(i) = -1, \quad (\text{C3})$$

$$g_{\text{other}}^{\text{behind}} > v_{\max}, \quad (\text{C4})$$

This uses the CWS rules of Chowdhury et al. [6]. This biases the asymmetric variant towards a populated slow lane and a sparser overtaking lane, matching the UK driving convention. In this report we focus on the asymmetric version as it provides a more realistic model

Heterogeneous traffic is modelled by a Bernoulli draw at injection: each vehicle is assigned slow type with probability f_{slow} (ceiling $v_{\text{max}}^{\text{slow}}$) and fast type otherwise (ceiling $v_{\text{max}}^{\text{fast}}$); the vehicle type is then fixed for the vehicle lifetime. The homogeneous case is recovered by setting $f_{\text{slow}} = 0$, so that every vehicle uses the same velocity ceiling $v_{\text{max}}^{(i)} = v_{\text{max}}^{\text{fast}}$. The two-lane extension also uses the same treatment for the lights as the single lane model does, where the red cell is added to the obstacle list for both lanes, so a single light will be applied to both lanes simultaneously.

3.7 Two-Dimensional Grid Extension and The Road Turning Rule

The two-dimensional grid extends the single lane state to an $N \times N$ network of perpendicular roads, on a torus (periodic boundaries in both dimensions).

$$\text{road}_e(t) \in \{-1, 0, 1, \dots, v_{\text{max}}\}^{N \times L}, \quad \text{road}_n(t) \in \{-1, 0, 1, \dots, v_{\text{max}}\}^{L \times N}, \quad (7)$$

where road_e holds the N west-to-east lanes and road_n holds the N south-to-north lanes. Each road is a one dimensional lattice of length L with the same NaSch dynamics described in the single lane case.

Intersections and traffic lights are evenly spaced along each row and column:

$$c_j = j D, \quad D = L/N, \quad j = 1, 2, \dots, N, \quad (8)$$

giving N^2 intersections and N^2 traffic lights in total.

Each time step proceeds in two stages: first are the four NaSch rules applied in a strict parallel update in each row and column. Secondly, any car sitting on an intersection cell turns onto the perpendicular road with probability p_{turn} , otherwise it continues straight.

3.8 Traffic Lights as Obstacles

Traffic lights are modelled as obstacles that can be either red or green. A red light is an obstacle that vehicles cannot pass through, while a green light allows vehicles to pass through without stopping.

A red light at cell c is added to the obstacle list used by the deceleration rule (3), so any approaching vehicle brakes to stop one cell short of c , exactly as it would be behind a stationary vehicle.

A green light is invisible to the dynamics, and cars can pass through it without stopping, so it is not added to the obstacle list and does not affect the gap g_i for any vehicle.

The N lights are placed at fixed positions along the road

$$c_i = \left\lfloor \frac{L i}{N+1} \right\rfloor, \quad i = 1, 2, \dots, N, \quad (9)$$

so that adjacent lights are separated by a distance of approximately $L/(N+1)$ cells.

Each light has three timing parameters: the cycle time T_{cycle} , the green phase duration T_{green} (with $0 < T_{\text{green}} < T_{\text{cycle}}$), and the per light offset parameter $\varphi_i \in \{0, 1, \dots, T_{\text{cycle}} - 1\}$. Together these determine the light's state at any time t through the phase counter:

$$\text{phase}_i(t) = (t + \varphi_i) \bmod T_{\text{cycle}}, \quad (10)$$

Light i is green when $\text{phase}_i(t) < T_{\text{green}}$ and red otherwise. Within each cycle, a green window of length T_{green} is followed by a red window of length $T_{\text{red}} = T_{\text{cycle}} - T_{\text{green}}$. The offset array $\{\varphi_i\}_{i=1}^N$ is the optimisation lever for the strategies studied later in the report:

- Setting $\varphi_i = 0$ for all i corresponds to the synchronised strategy, where every light switches in phase
- Setting $\varphi_i = (i-1)T_{\text{cycle}}/N$ is a green wave strategy, where lights are staggered so each car reaches the light as it turns green
- Setting $\varphi_i \sim \text{Uniform}(0, T_{\text{cycle}})$ is a random offset strategy.

In the two lane case the same red cell is in the obstacle list for both lanes, so a single light will be applied to both lanes simultaneously.

On the two-dimensional grid, placement differs. The N^2 lights sit at intersection cells. There is one light per intersection, which forces exclusion between the horizontal and vertical traffic through a per-intersection phase counter with cycle length $2T$:

$$\text{phase}_{ij}(t) = (t + \delta_{ij}) \bmod 2T, \quad (11)$$

where δ_{ij} is the offset of light (i, j) . The lights are green when $\text{phase}_{ij}(t) < T$ and red otherwise. Horizontal cars pass when the light is green, and vertical cars pass when the light is red, so the light forces an alternation between horizontal and vertical traffic at each intersection. The same three strategies are applied but now over the two-dimensional offset array $\{\delta_{ij}\}$.

3.9 Boundary Conditions

Two boundary regimes are used in this report but not simultaneously. The periodic regime is used so that a closed system at a fixed density ρ gives access to the steady-state flow $J(\rho)$. The open regime is used for every experiment involving traffic lights or sensitivity sweeps, where throughput is a quantity we are analysing. The city grid is always periodic in both dimensions (a torus). No open boundary. No open-boundary version of the model is explored.

Periodic boundaries take cell indices modulo L , so any vehicle whose updated position would exceed $L - 1$ wraps back to the start of the lattice; the total number of vehicles n_{cars} is conserved. The system has density

$$\rho = \frac{n_{\text{cars}}}{n_{\text{lanes}} L}, \quad (12)$$

where $n_{\text{lanes}} \in \{1, 2\}$.

The periodic-boundary through-flow observable is the time-averaged mean velocity multiplied by density,

$$J = \langle v \rangle \rho. \quad (13)$$

Open boundaries have a finite length: at every step, a candidate vehicle is injected at cell 0 in each lane with probability p_{in} , provided that cell is empty. Newly injected vehicles enter with velocity zero and accelerate from there with the NaSch rules. Any vehicle whose updated position satisfies $i + v \geq L$ exits the lattice and increments the outflow counter n_{out} .

The open-boundary through-flow observable is the per-step exit rate:

$$J = \frac{n_{\text{out}}}{T_{\text{measure}}}, \quad (14)$$

In the open-boundary regime, the lattice is initialised empty (all cells set to -1), and traffic builds up from the injection point over the warmup period. In the periodic-boundary regime the n_{cars} vehicles are placed at distinct cells drawn uniformly without replacement in both lanes (if two-lane), with initial velocities drawn uniformly from $\{0, 1, \dots, v_{\text{max}}\}$. In both regimes, the system has a warmup period before any flow is recorded, discarding the initial dynamics and allowing the system to reach a steady state before measurements are taken.

4 Numerical Method and Validation

The cellular-automaton model is simulated so that at every timestep a single snapshot of the lattice state is taken, the four NaSch rules of equations (2)-(5) are compared against the snapshot for all vehicles in parallel, the resulting positions are then written back. This allows for the parallel-update stated in Section 3.1 and does not give an advantage to some vehicles. The braking rule refers to the obstacle list which include other vehicles and red lights Section 3.8, the list is then built once per step with `numpy.searchsorted` for $O(\log n)$ neighbour lookups.

Flow measurements use two phases: the first T_{warmup} steps are used so the system can go to its steady state, and the next T_{measure} steps are used to record J via equation (13) for periodic boundaries or equation (14) for open boundaries. Each run is repeated for n_{seeds} independent random seeds; the mean and ± 1 SD across seeds are reported as the average with the shaded band in every figure. Sweep results are cached to `.npz` files so that the figure set regenerates from the same deterministic numbers on every run.

The implementation was validated against two checks. First a low density of the fundamental diagram of Figure 3 reproduces the analytic free-flow slope $v_{\text{free}} = v_{\text{max}}p \approx 4.5$, and secondly a regression test that make sure that the two lane code $n_{\text{lanes}} = 1$, $p_{\text{chg}} = 0$ reproduces the same diagram as the single lane code.

5 Baseline Results

5.1 No-Lights Baseline

Before studying the effects of traffic lights and the addition of a second lane, we first establish a baseline. In this subsection we will look at the one dimensional lattice with no traffic lights.

5.1.1 Baseline Space-Time Evolution

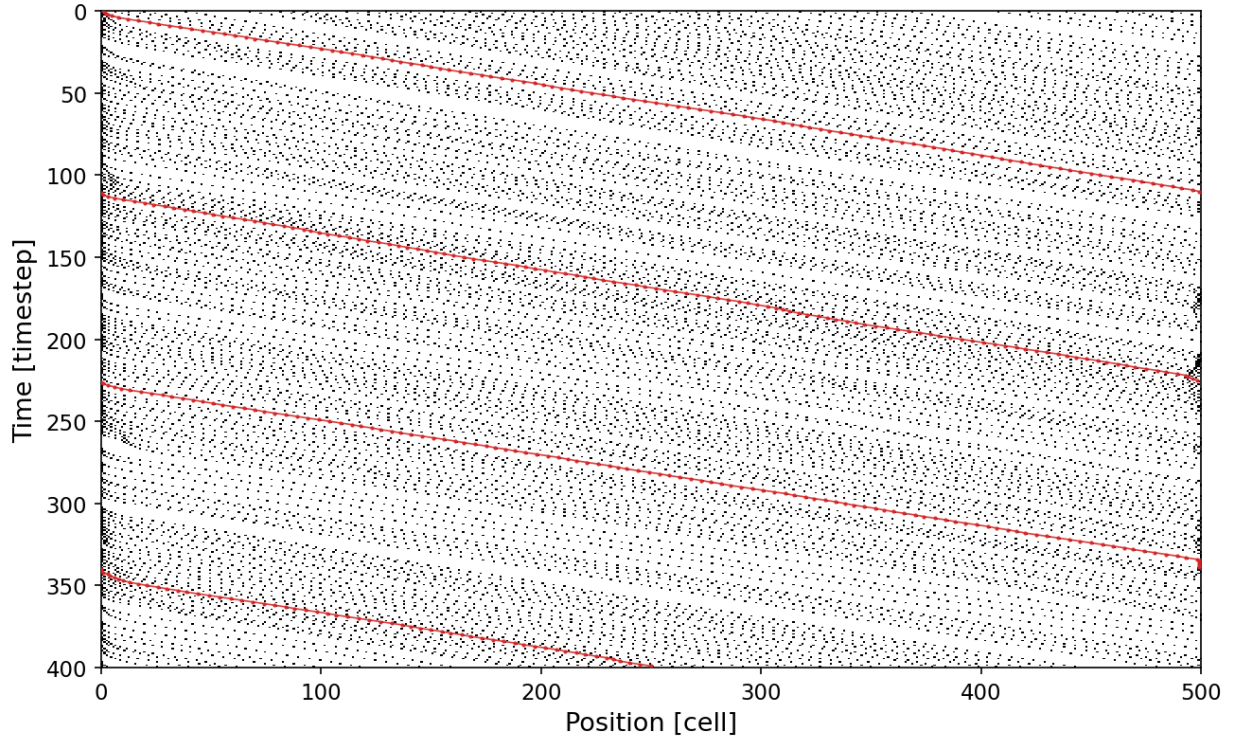


Figure 2: Single-lane space-time diagram at $L = 500$, $v_{\max} = 5$, $p_{\text{rand}} = 0.3$, $p_{\text{in}} = 0.5$, no traffic lights, recorded over $T = 400$ steps after $T_{\text{warmup}} = 500$. Each black pixel is an occupied cell; one car is tagged in red so its trajectory can be traced through the diagram.

In Figure 2 the spatio-temporal dynamics of the one dimensional system, without traffic lights we can see the base traffic flow dynamics without any traffic lights or lane changes. We can see the trajectory of a vehicle with the red line. Additionally, we can see there are darker regions which is due to the random deceleration rule which allows for stop and go waves to form.

5.1.2 Baseline Fundamental Diagram

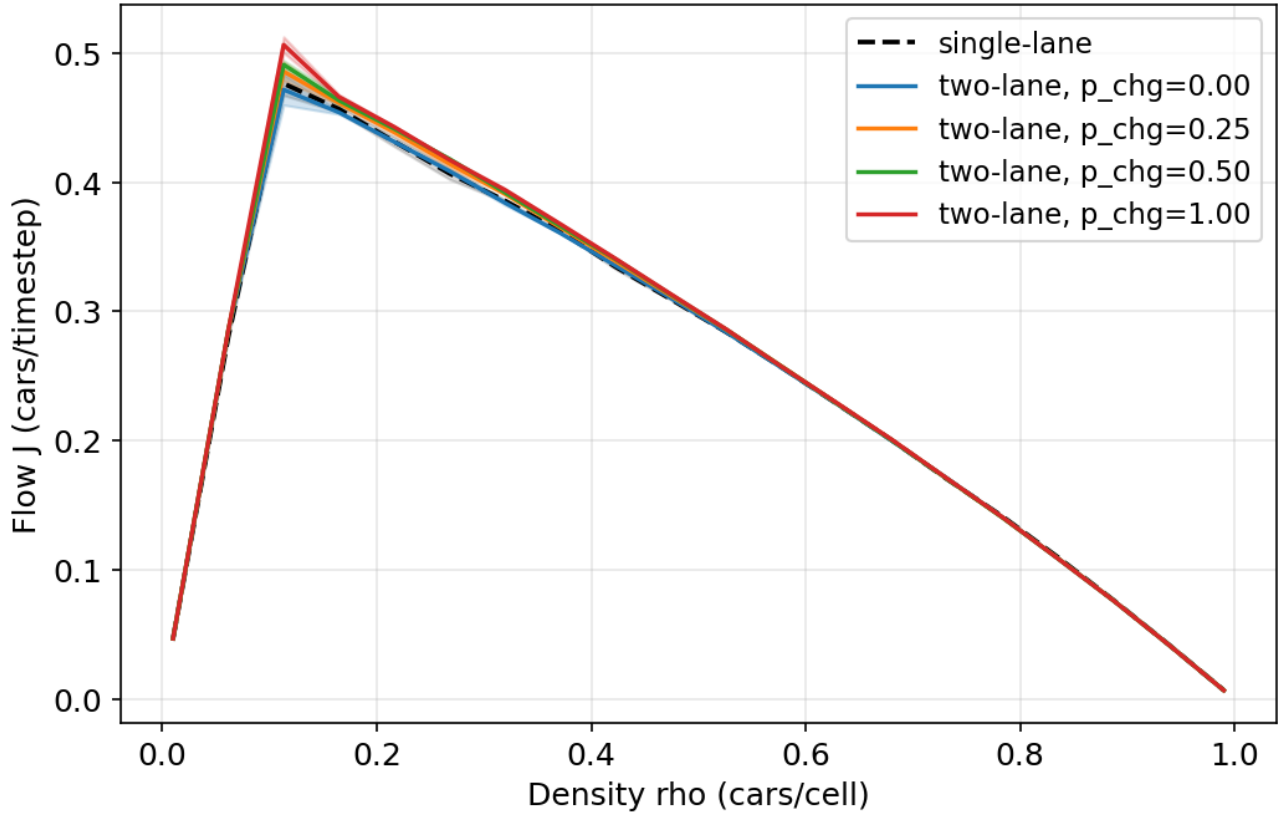


Figure 3: Fundamental diagram $J(\rho)$ for the single-lane (dashed) and two-lane NaSch model under periodic boundaries, $L = 500$, $v_{\max} = 5$, $p_{\text{rand}} = 0.3$. Two-lane curves are tested for four lane-change probabilities $p_{\text{chg}} \in \{0, 0.25, 0.5, 1.0\}$. Each point is the mean of $n_{\text{seeds}} = 3$ independent runs after $T_{\text{warmup}} = 2000$ discarded steps and $T_{\text{measure}} = 5000$ measurement steps.

In Figure 3 we can see the fundamental diagram for the single-lane and two-lane NaSch model under periodic boundaries. We can see that the trend across both the single-lane and two-lane is the same with a linear increase up to $\rho \approx 0.15$. After this point the flow decreases gradually getting closer to zero as ρ approaches 1. This shows that although more cars can be on the road with two lanes, flow isn't necessarily higher. Additionally we can see with more cars there is more chance for congestion due to the random deceleration rule.

5.2 Single-Lane Lights

We now place N equidistant traffic lights on the single lane system and study how the cycle parameters reshape the traffic flow J . We will in particular explore three different strategies for coordinating the traffic lights: in phase (all lights switch together), green wave (lights are offset by T_{cycle}/N , so that they are green as soon as the car reaches the light), and random (lights have independent uniform offsets).

5.2.1 Phase Offset and Green Wave

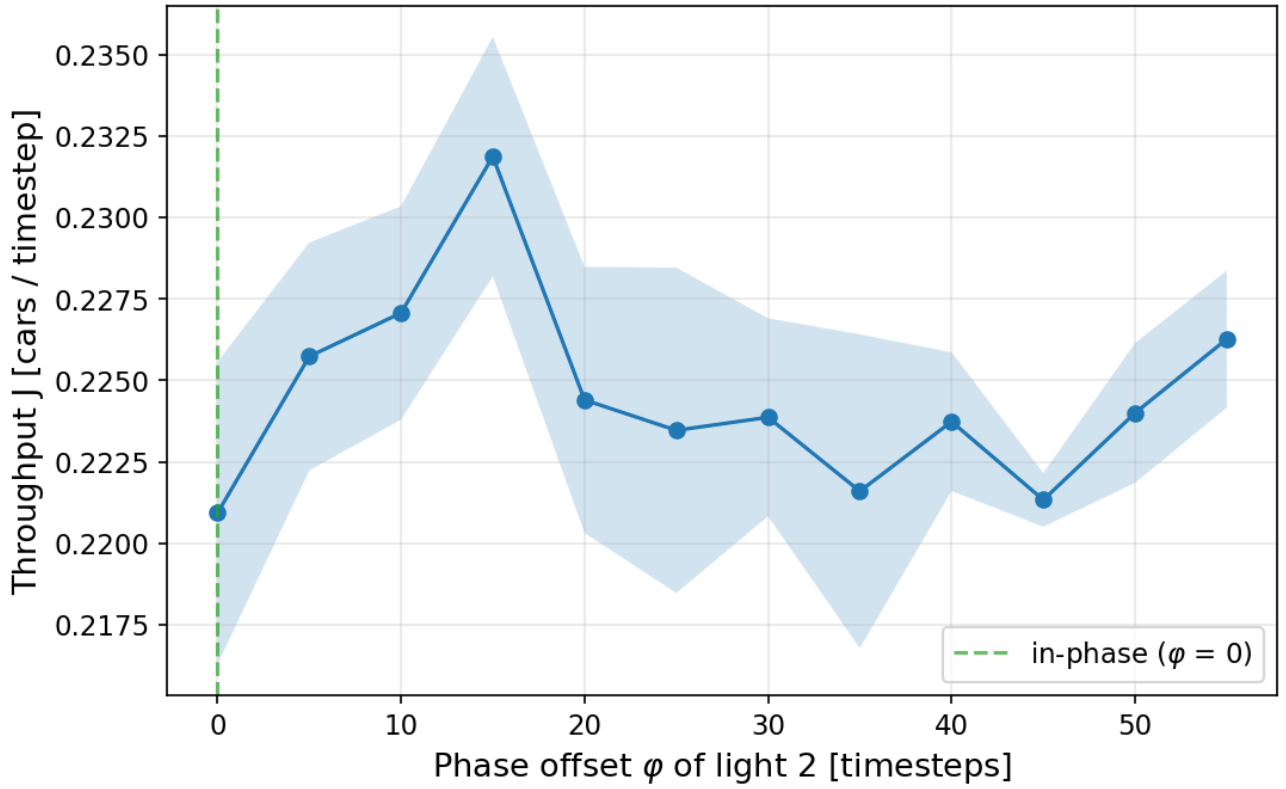


Figure 4: Single-lane J versus phase offset ϕ at $N = 2$, $T_{\text{cycle}} = 60$, $T_{\text{green}} = 30$, $p_{\text{in}} = 0.4$. Bands are ± 1 SD over $n_{\text{seeds}} = 3$ runs of $T_{\text{measure}} = 2000$ steps after $T_{\text{warmup}} = 1000$.

In Figure 4, we run a phase offset sweep for the single-lane system with $N = 2$ lights and $T_{\text{cycle}} = 60$. We see that the throughput has no clear trend as we increase the phase offset, however we can see that the optimal phase offset is around $\phi = 15$ which could be the green wave strategy, which is consistent with our analytic prediction of $\phi = T_{\text{cycle}}/N = 30$. This suggests that rather than all the lights switching together, it is better to have them switch in a way that allows cars to pass through more lights without stopping.

5.2.2 Strategy Comparison against various Light Counts

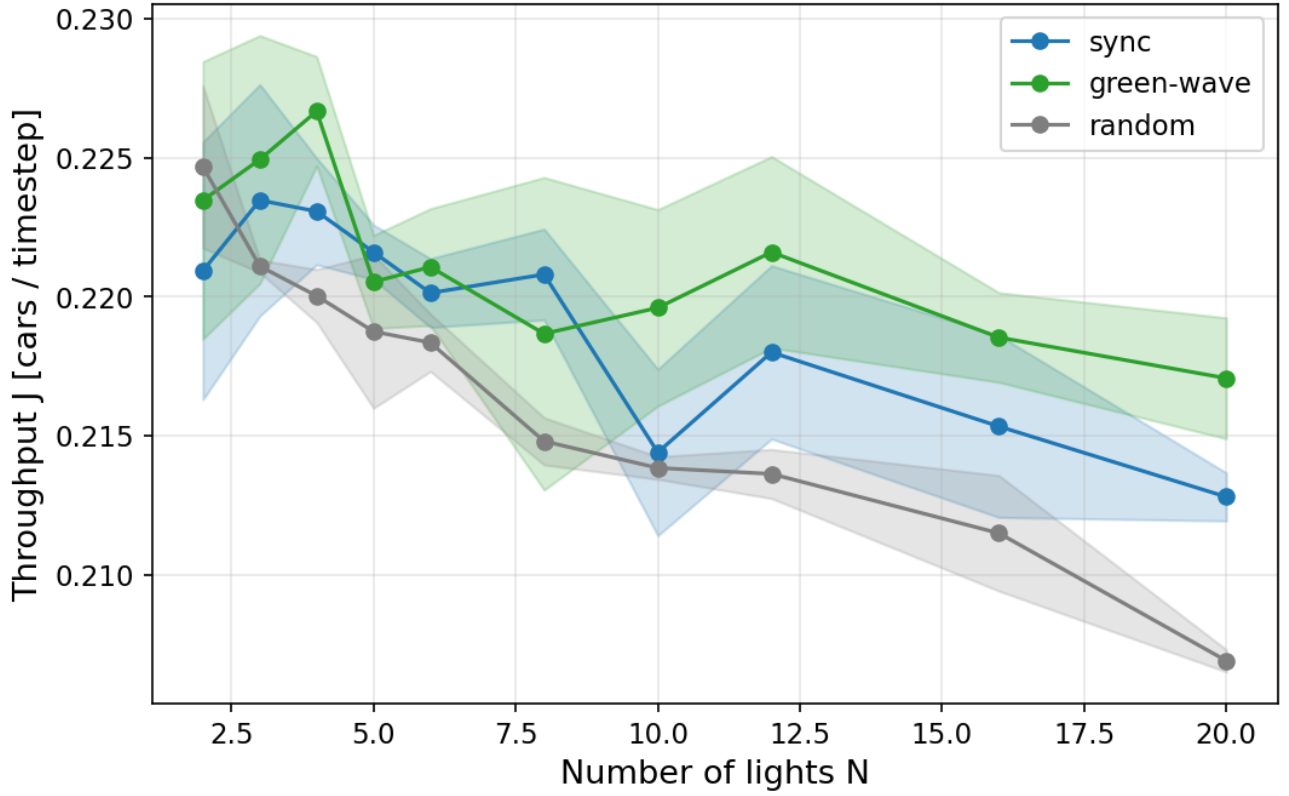


Figure 5: Single-lane J versus N under three coordination strategies (sync, green-wave, random) at $T_{\text{cycle}} = 60$, $T_{\text{green}} = 30$, $p_{\text{in}} = 0.4$. Bands are ± 1 SD over $n_{\text{seeds}} = 3$ runs, for random offset phases are additionally averaged over five independent uniform offset values.

As we test our different strategies in Figure 5, we can see that at low light counts $N < 6$ all strategies perform similarly. However, as we increase the number of lights, the green wave strategy starts to consistently outperform the other two strategies. Whilst random offset performs the worst, becoming more and more unreliable as the number of lights increases.

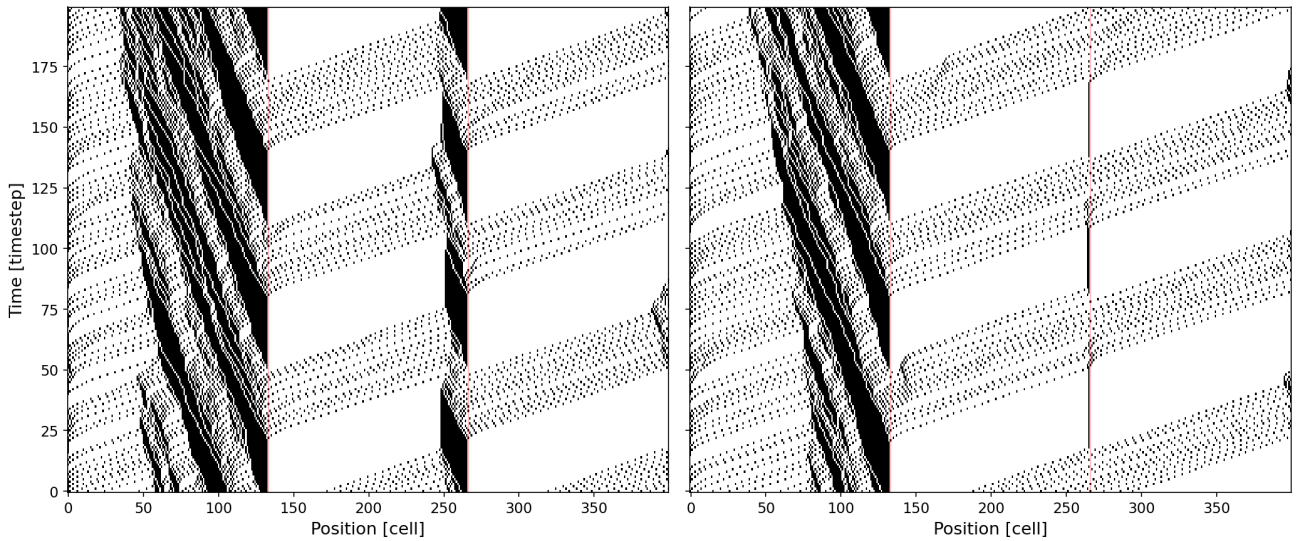


Figure 6: Space-time diagrams at $N = 2$, $T_{\text{cycle}} = 60$, comparing in-phase ($\varphi = 0$, left) and green-wave ($\varphi = 32$, right) coordination. $L = 400$, $v_{\text{max}} = 5$, $p_{\text{rand}} = 0.3$, $p_{\text{in}} = 0.4$. Red vertical lines mark the light positions at $L/3$ and $2L/3$.

In Figure 6, we compare our two best strategies. Here we can see that our green wave strategy allows for more cars to pass through and not be halted at the traffic light. We also see that when cars are stuck at the traffic light more of them get through in contrast to the synchronised strategy highlighting green waves ability to quickly get rid of any build up that has occurred.

5.3 Two-Lane Lights

We now extend our traffic light model to an asymmetric two-lane setting, where lane changes are allowed but the traffic lights control both lanes and strategies are based on the slow lane. The left lane is preferred so cars will try to move back into it as soon as possible when overtaking.

5.3.1 Asymmetric CWS and Lane Bias

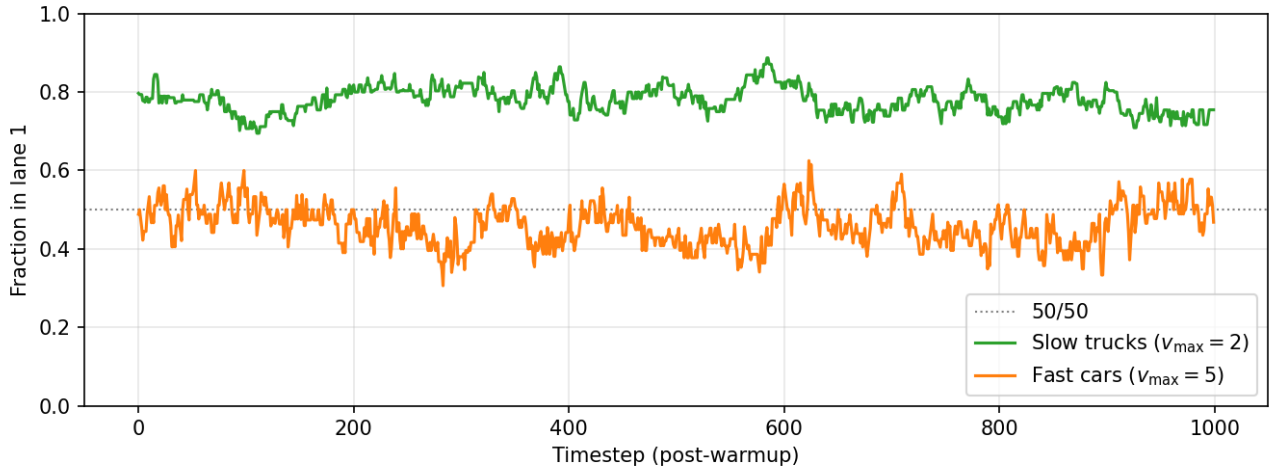


Figure 7: Lane-1 occupancy under the asymmetric system at $L = 1000$, $N = 2$ synchronised lights, $T_{\text{cycle}} = 60$, $T_{\text{green}} = 30$, $p_{\text{rand}} = 0.1$, $p_{\text{in}} = 0.4$, $p_{\text{chg}} = 1.0$, $f_{\text{slow}} = 0.5$, $v_{\text{max}}^{\text{slow}} = 2$, $v_{\text{max}}^{\text{fast}} = 5$, recorded over 1000 post-warmup steps. Dotted grey line marks the 50/50 reference. Slow vehicles are in orange, whilst fast vehicles are in blue

In Figure 7, we can see the lane occupancy for the asymmetric system. We can see that cars tend to bias heavily towards the slow lane within the UK, especially slow trucks, which have been much less likely to be in a situation where they feel a need to overtake hence why around $\approx 80\%$ of trucks stay in lane 1. Faster vehicles are more evenly split between both lanes, with the percentage being slightly lower than 50% majority of the time. This behaviour is also realistic when applied to UK traffic laws as drivers tend to bias the left lane and vans are much less prone to overtaking due to their lower speed limits.

5.3.2 Phase Offset and Green Wave

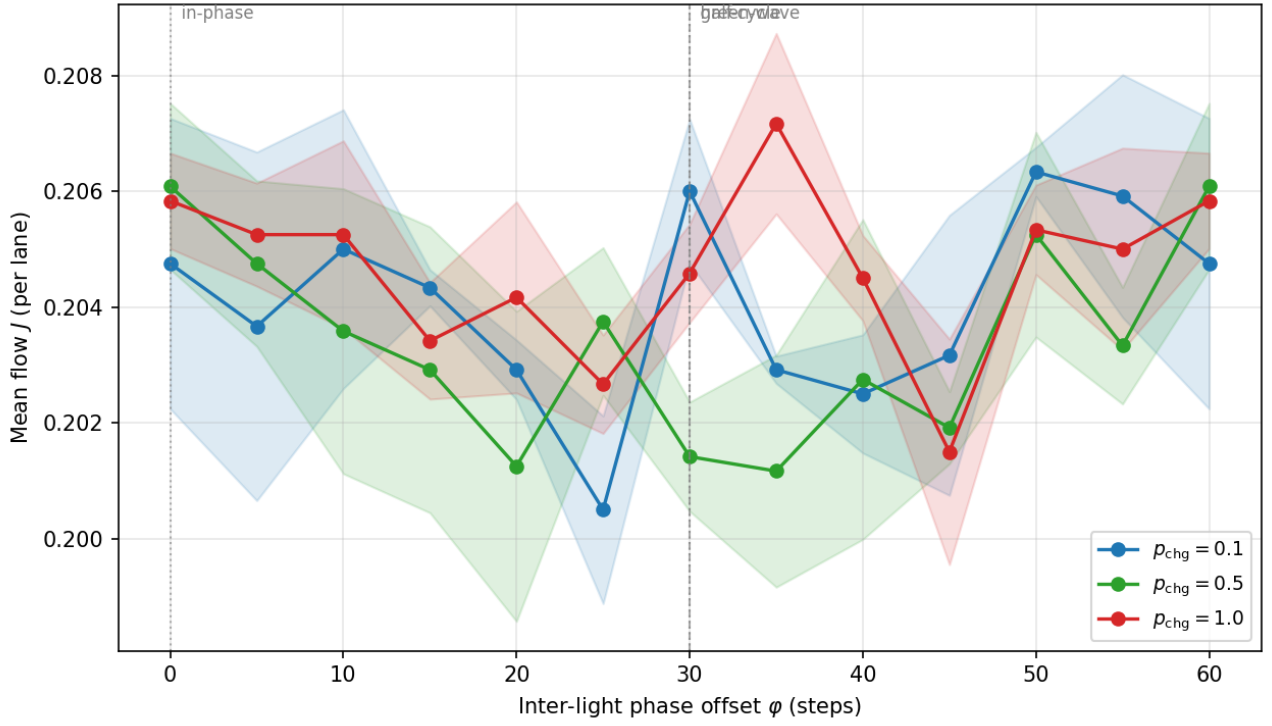


Figure 8: Asymmetric CWS: J versus phase offset ϕ at $N = 2$, $T_{\text{cycle}} = 60$, $T_{\text{green}} = 30$, $p_{\text{in}} = 0.4$, $f_{\text{slow}} = 0.5$, for three lane-change probabilities $p_{\text{chg}} \in \{0.1, 0.5, 1.0\}$. Grey verticals mark in-phase ($\phi = 0$), analytic green-wave ($\phi = T_{\text{cycle}}/N = 30$), and half-cycle. Bands are ± 1 SD over $n_{\text{seeds}} = 3$ runs of $T_{\text{measure}} = 2000$ steps after $T_{\text{warmup}} = 1000$.

In Figure 8, we can see the phase offset sweep with $N = 2$ lights and $T_{\text{cycle}} = 60$ for the asymmetric system. We see that even when we have various lane change probabilities, there is no clear difference in the effect of changing lane highlighting that lane changing may not have a significant effect on the optimal traffic strategy.

5.3.3 Light Count

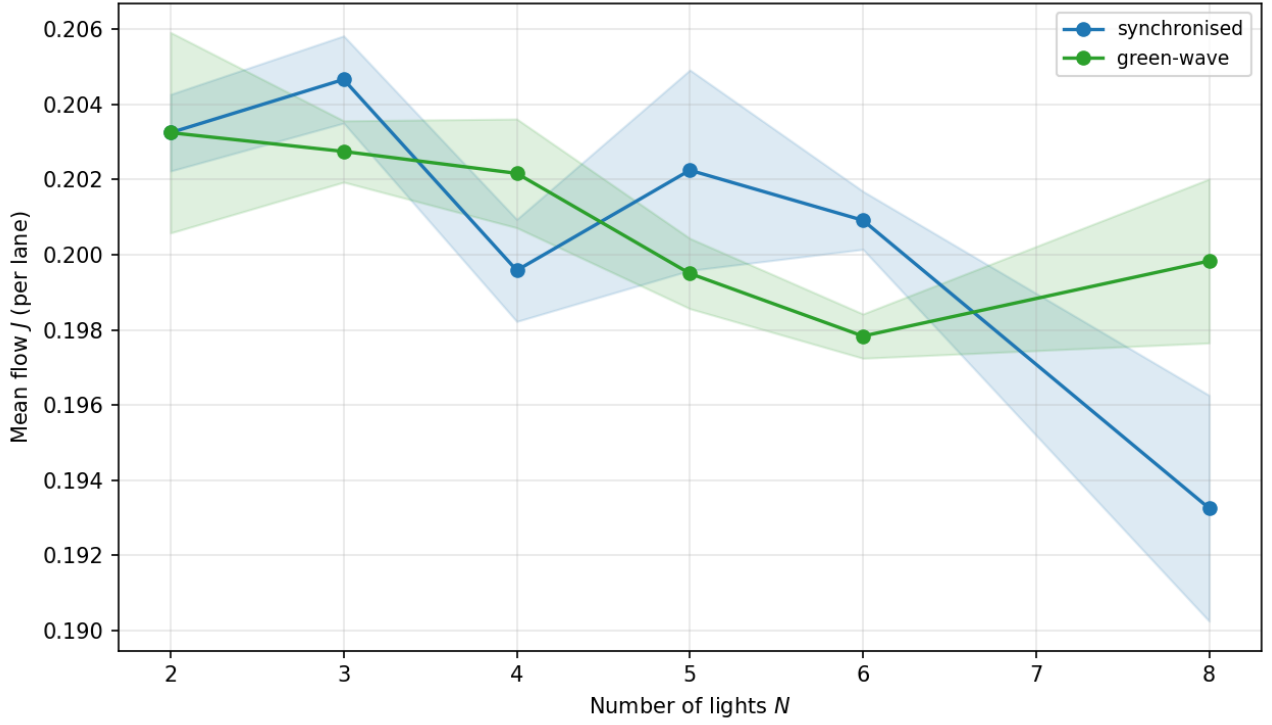


Figure 9: Asymmetric CWS: J versus light count N under sync and green-wave coordination, at $L = 1000$, $T_{\text{cycle}} = 60$, $T_{\text{green}} = 30$, $p_{\text{in}} = 0.4$, $p_{\text{chg}} = 1.0$, $f_{\text{slow}} = 0.5$. Bands are ± 1 SD over $n_{\text{seeds}} = 3$ runs.

In Figure 9, we can see the effects of the two best strategies as we increase the number of lights. We can see that the synchronised strategy degrades as we increase the number of lights, following the same trend as in the single lane system, whereas with the green wave strategy it doesn't degrade as much suggesting that it may be more scalable for larger systems.

5.4 Two-Dimensional City Grid

The single lane and lane changing extensions are useful for understanding the effects of traffic lights in a simple setting, but real-world traffic networks are more complex and go beyond a one-dimensional system. To explore the effects of traffic light coordination in a more realistic setting, we now study our $N \times N$ city grid model, which consists of two-dimensional grid of one-lane roads with traffic lights at every intersection.

5.4.1 Mechanism Visualisation

G4 — Grid snapshots: sync (top) vs wave (bottom) at two times ($N = 4$, $T = 20$, $\rho = 0.1$, $D = 50$, green dot = east-green; red dot = north-green)

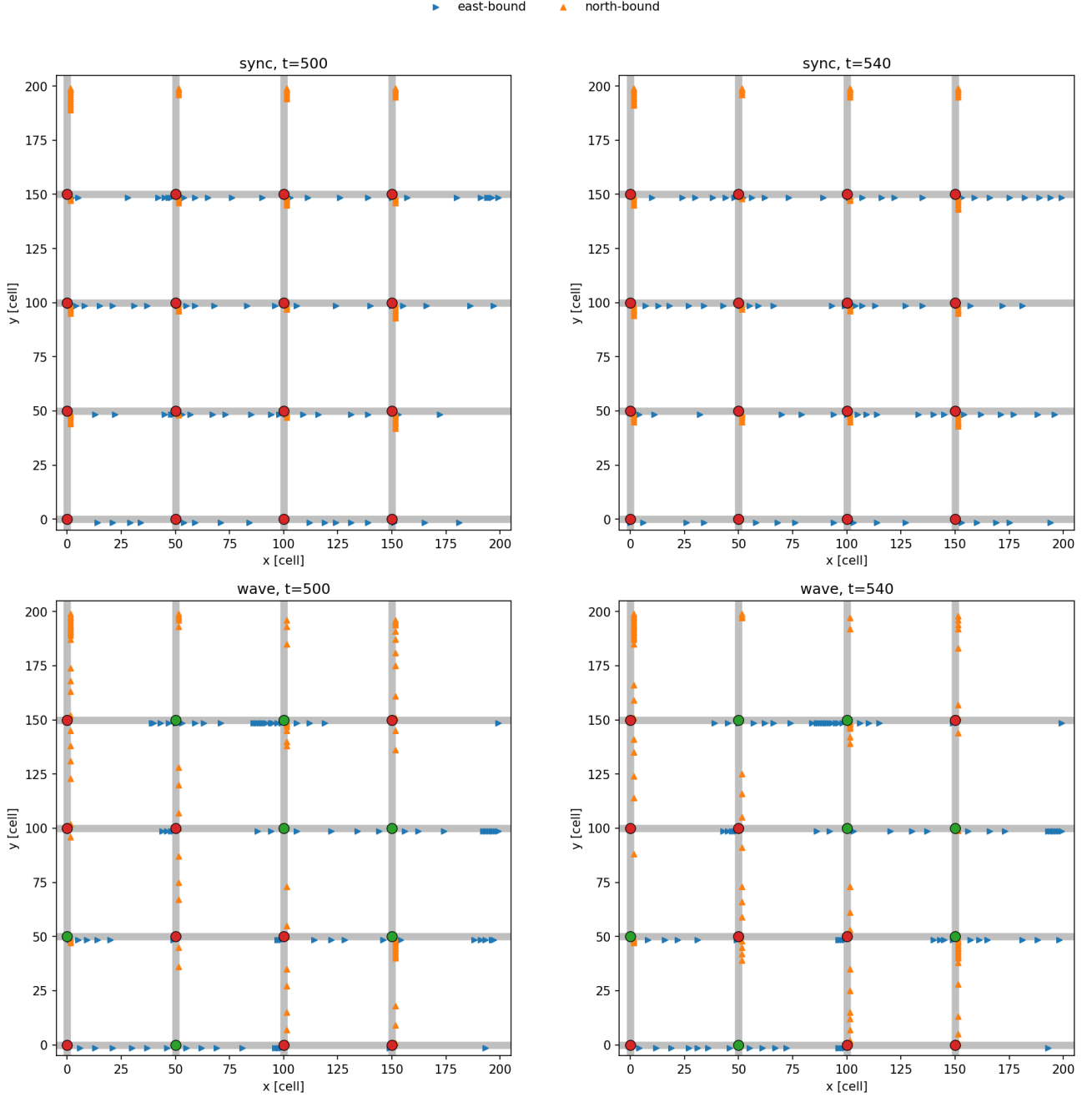


Figure 10: Grid snapshots: sync (top row) vs wave (bottom row) at two timesteps separated by one full cycle ($2T = 40$). Parameters: $N = 4$, $T = 20$, $\rho = 0.10$, $D = 50$, $v_{\max} = 5$, $p_{\text{rand}} = 0.3$, single seed. East-bound cars are right-pointing triangles on row lines; north-bound cars are up-pointing triangles on column lines; intersection discs are coloured green when east has the green phase and red when north does.

Here in Figure 10, we can see the snapshots of the grid system under the synchronised and green wave strategies. We can see that with the synchronised strategy, all the traffic lights are red at the same time, which causes a lot of congestion as cars have to wait for all the lights to turn green at the same time. On the other hand, with the green wave strategy, we can see a pattern of green and red lights that allows cars to pass through more smoothly without having to stop at every light. This visual evidence supports our earlier findings that the green wave strategy is more effective at improving traffic flow compared to the synchronised strategy.

5.4.2 Grid Baseline and Fundamental Diagram

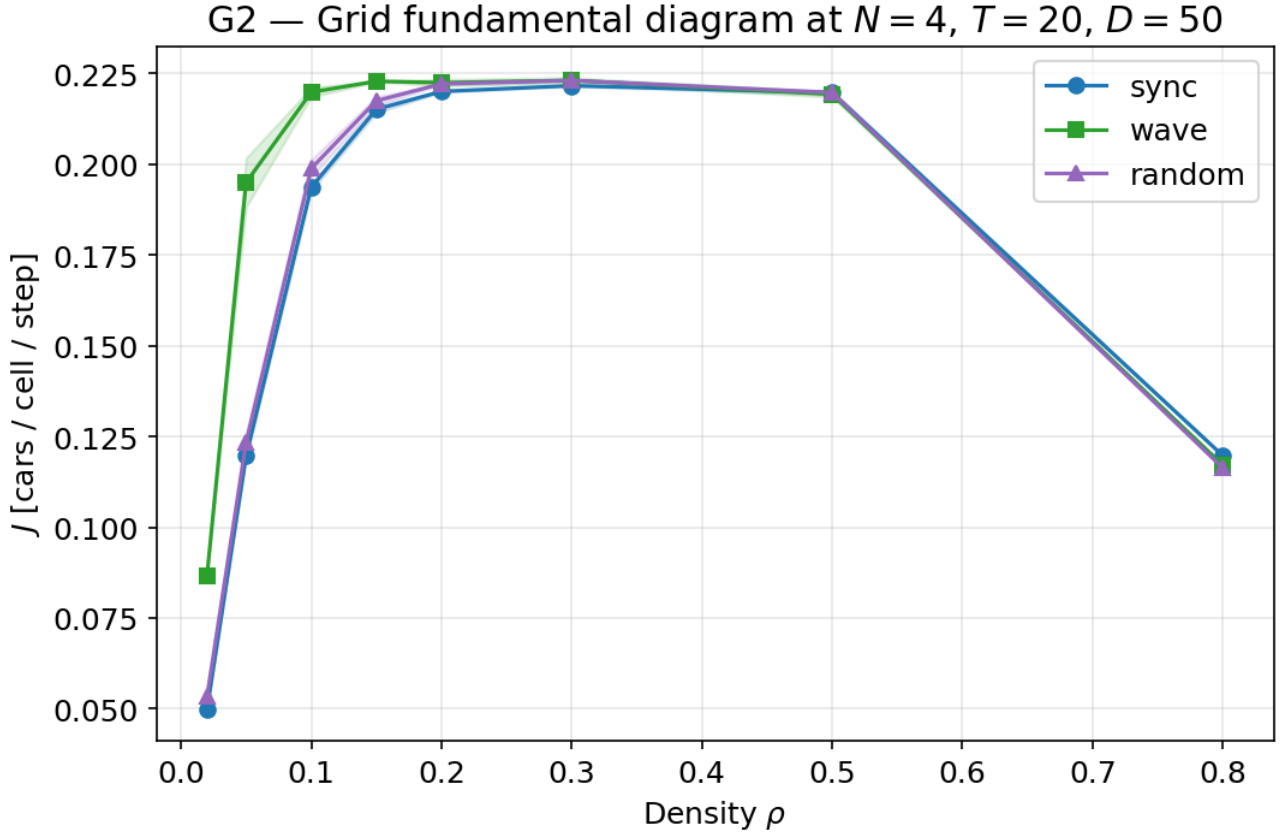


Figure 11: Grid fundamental diagram $J(\rho)$ for $N \times N$ city grid at $N = 4, T = 20, D = 50$, for the three offset strategies (sync, wave, random). Periodic boundaries; density is conserved. Parameters: $v_{\max} = 5, p_{\text{rand}} = 0.3, T_{\text{warmup}} = 1000, T_{\text{measure}} = 3000, n_{\text{seeds}} = 3$. Shaded bands are ± 1 SD over seeds.

In Figure 11, we can see the fundamental diagram for the $N \times N$ city grid at $N = 4$ for the three offset strategies. Unlike the fundamental diagrams in the previous sections we do not have a linear increase but rather a more gradual increase up to a peak at around $\rho \approx 0.15 - 0.20$ after which the flow decreases gradually afterwards. We can see the green wave strategy converges the fastest in contrast to the other two, we can also see that the trend in synced lights and the random offset strategies are almost identical, indicating that they have similar behaviour in a 2D system. At $\rho = 0.5$ we can see a drop in all of them highlighting that no matter what strategy, all will lead to the same outcome when car density is high enough.

5.4.3 Strategies across increasing N

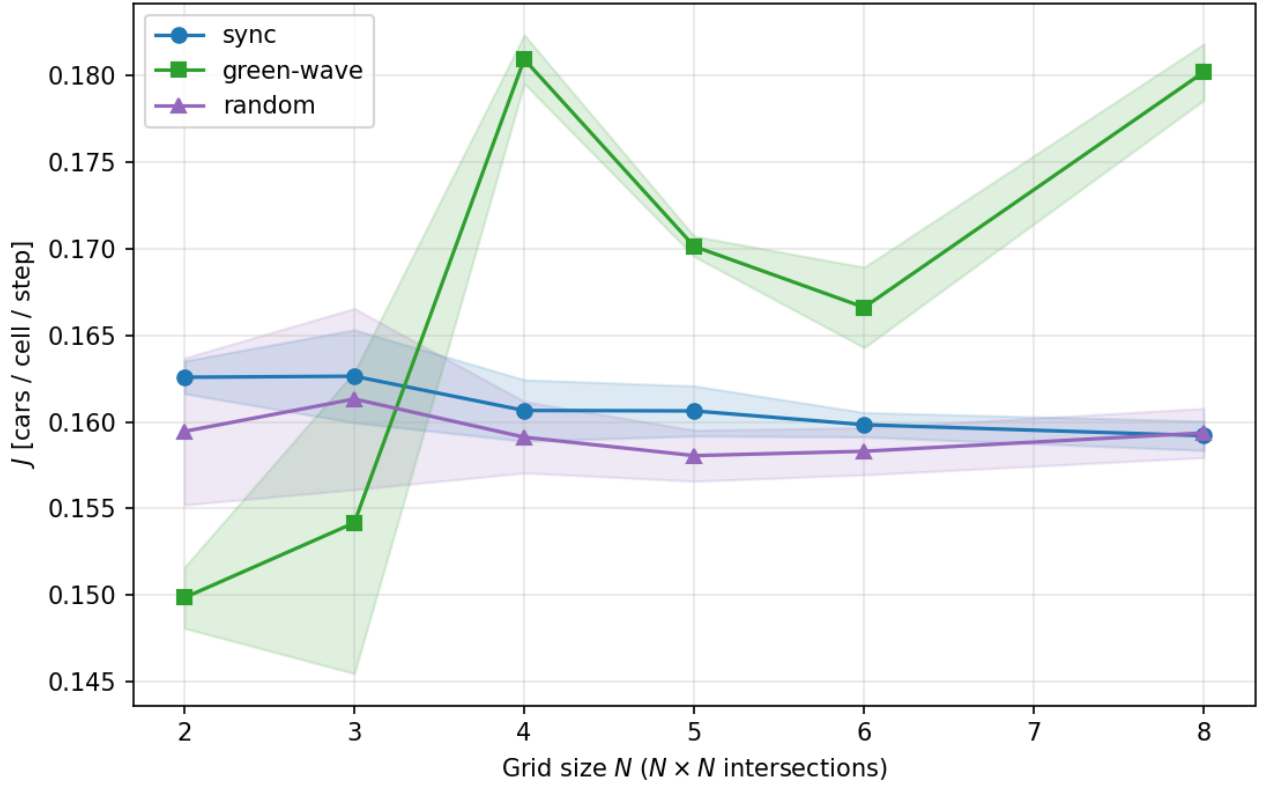


Figure 12: Grid through-flow J versus grid size N for the three coordination strategies (sync, green-wave, random) with turning enabled at $p_{\text{turn}} = 0.5$. Parameters: $T = 20$, $\rho = 0.10$, $D = 50$, $v_{\text{max}} = 5$, $p_{\text{rand}} = 0.3$, $T_{\text{warmup}} = 1000$, $T_{\text{measure}} = 3000$, $n_{\text{seeds}} = 3$. Shaded bands are ± 1 SD across seeds.

In Figure 12, we can see the effects of the three strategies as we increase the grid size N with a turning rate of $p_{\text{turn}} = 0.5$. We can see that at low grid sizes with $N \leq 3$ that green wave does not have an advantage over the other strategies but eventually it dominates significantly over the other two strategies suggesting that it is the more scalable option, the difference is also more noticeable than the other approaches. On the other hand we see that synced lights and random offset strategies perform similarly across all grid sizes, with no significant change in flow as we increase the grid size. We can see that the benefits gained from green wave are only really seen as we scale the model up.

5.4.4 Turning Cars Extension

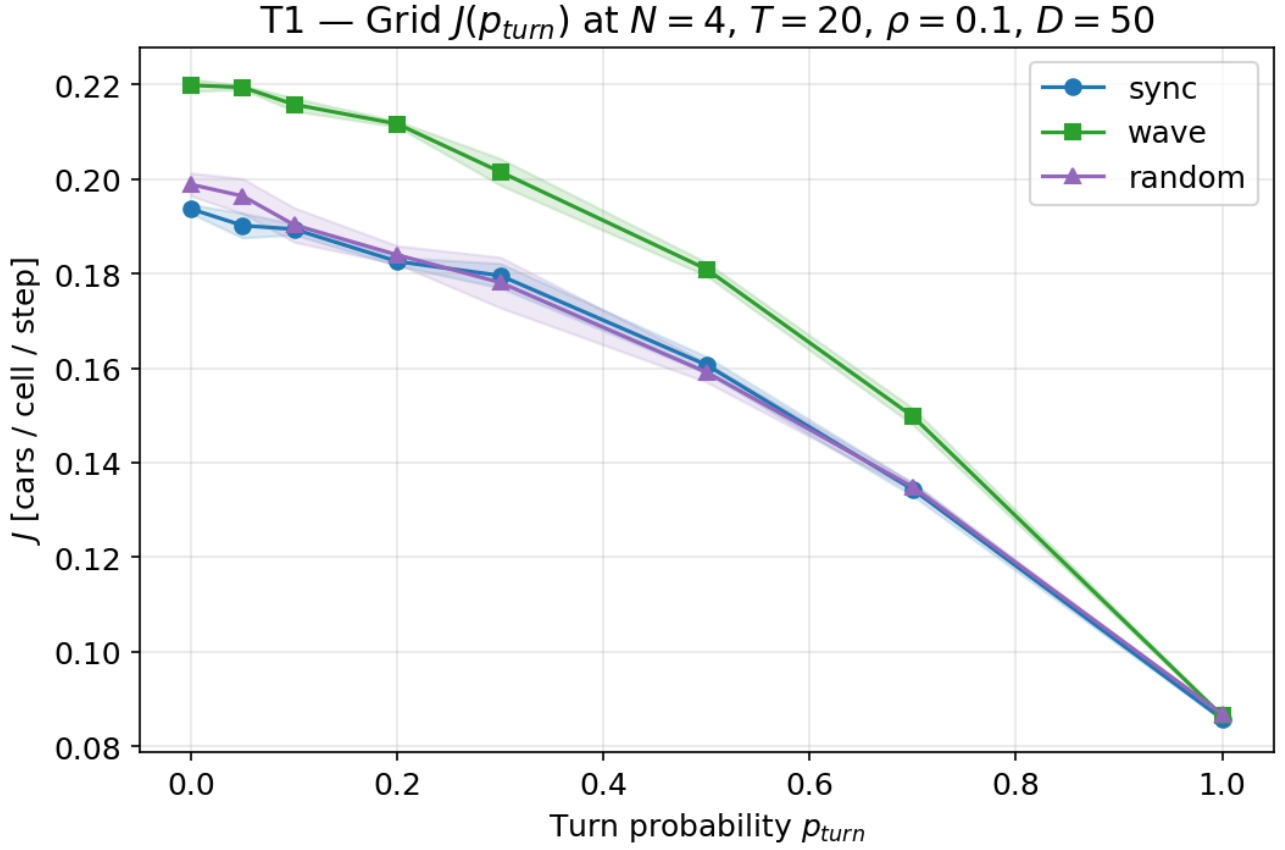


Figure 13: Grid through-flow J versus per-intersection turn probability p_{turn} for the three offset strategies, at fixed $N = 4, T = 20, \rho = 0.10, D = 50, v_{max} = 5, p_{rand} = 0.3, T_{warmup} = 1000, T_{measure} = 3000, n_{seeds} = 3$. Each successful turn resets the car’s velocity to zero.

In Figure 13, we can see the effects of increasing the turning probability p_{turn} on the three strategies. We can see that although green wave is the best strategy initially it decreases as we increase the turning probability, however it still maintains an advantage over the other two strategies up until the car turns at every intersection. This suggest that optimisation strategies may not be able to accurately account for the unpredictability of human drivers, and that the benefits of green wave may be less significant in real world settings where drivers have more freedom to change direction.

6 Sensitivity Analysis

6.1 Methodology

To assess sensitivity, one parameter is varied at a time while all others are held fixed. For each parameter p , the flow is measured at up to two perturbed values, $0.9p$ and $1.1p$, to estimate the sensitivity coefficient S_i for that parameter. We simulate up to 2000 timestamps, $n_{\text{seeds}} = 2$. Here we use the equilibrium flow J^* as the performance metric, which is the average flow after a warmup period of T_{warmup} steps, measured over a subsequent period of T_{measure} steps, and averaged across n_{seeds} random seeds to reduce noise.

Each parameter is varied by $\pm 10\%$, giving values $0.9p$ and $1.1p$. We had also established baseline values to support our analysis. We will estimate the sensitivity coefficients using the central finite difference formula:

$$S_i = \frac{J^*(1.1p_i) - J^*(0.9p_i)}{0.2 J^*_{\text{baseline}}} \quad (15)$$

These coefficients give us an estimate of how strongly long-term traffic flow responds to changes in each parameter, with larger absolute values indicating greater change.

6.2 Tornado Plot Ranking

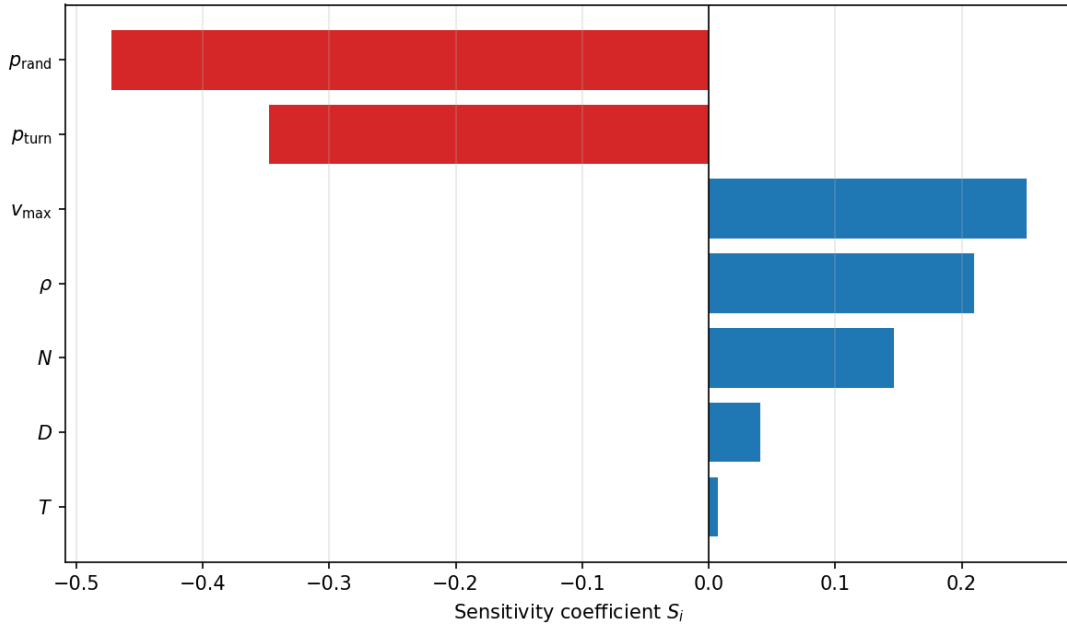
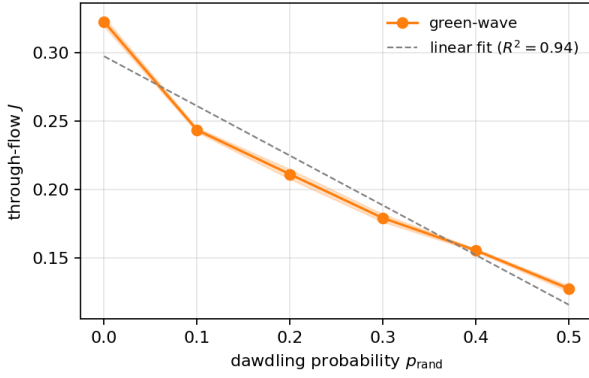


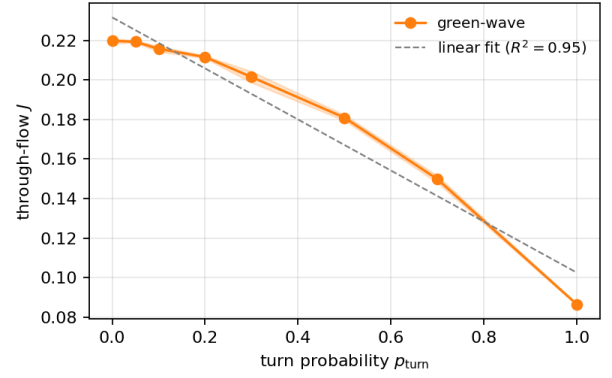
Figure 14: One-at-a-time sensitivity ranking of seven parameters around the 2D-grid-with-turning baseline under the green-wave coordination strategy.

In the tornado plot, Figure 14, we can see that the most sensitive parameter is the dawdling probability p_{rand} , followed by p_{turn} , which are the two parameters which focus on what we cannot control as much as the other ones in real life showing that drivers may be the ones causing the most congestion rather than the actual traffic policy. Additionally we can see that raising the speed limit up, improves throughput up to v_{max} beyond which the green-wave strategy becomes less effective. We can also see that the density of cars ρ is also a sensitive parameter, which makes sense up to a certain point since if more cars are in the system more cars will get through it, but if too many cars are in the system it will get congested and the flow will decrease. The other parameters have a much smaller sensitivity coefficient, which suggests that they may not be as important to focus on for improving traffic flow under the green-wave strategy.

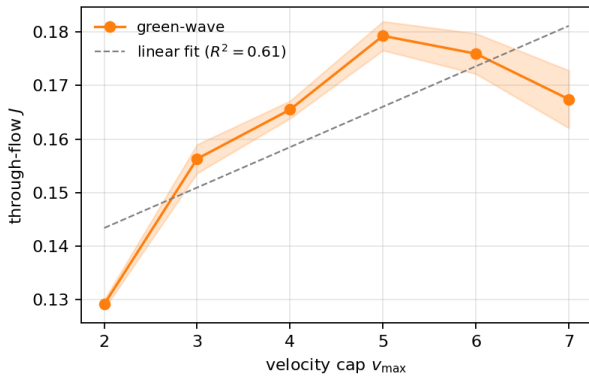
6.3 Single-Parameter Response Curves



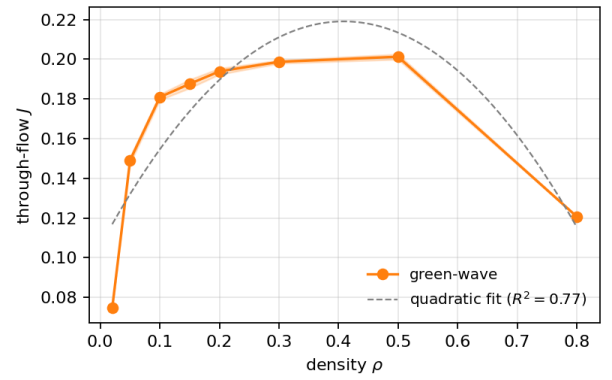
(a) $J(p_{\text{rand}})$ at $\rho = 0.10$, $N = 4$, $p_{\text{turn}} = 0.5$.



(b) $J(p_{\text{turn}})$ at $T = 20$, $\rho = 0.10$, $N = 4$.



(c) $J(v_{\text{max}})$ at $\rho = 0.10$, $N = 4$, $p_{\text{turn}} = 0.5$.



(d) $J(\rho)$ at $T = 20$, $N = 4$, $p_{\text{turn}} = 0.5$.

Figure 15: Single-parameter response of through-flow J on the 2D-grid-with-turning model under the green-wave coordination strategy for the four highest parameters from Figure 14: p_{rand} , p_{turn} , v_{max} , and ρ . Shaded bands are ± 1 SD over seeds.

In Figure 15, we can see the response curves for the four most sensitive parameters from the tornado plot. In panel (a), we see that as p_{rand} increases, the flow J decreases significantly, which indicates that higher dawdling rates leads to more congestion. In panel (b), we observe that increasing p_{turn} also reduces flow, but the effect is not as significant as when p_{rand} is increased. Panel (c) shows that increasing the speed limit v_{max} increases flow up to a certain point, after which the effect plateaus and slightly decreases. Finally, panel (d) illustrates that as density ρ increases, flow initially increases but then sharply decreases after a certain point, this also resembles the fundamental diagram of traffic flow. We can see also that (a-c) have a linear response, whilst (d) has a quadratic response. The magnitude of the slope on (a) is also large at -0.36 than the other parameters, which aligns with the observation that p_{rand} has the highest sensitivity coefficient.

7 Conclusion

7.1 Key Findings

Across all contexts we can see that the green wave strategy consistently outperforms the other two strategies, especially as we increase the number of lights or grid size. We can also see that when traffic dynamics are more complex that the effects of synced and randomised traffic light strategies are almost identical, suggesting that they have similar behaviour in more complex systems.

Besides identifying the best strategy, we also identified a range of patterns across the models. Firstly, we observed that the flow J scaled with the grid size N under the green wave strategy, on the other hand synchronised and randomised strategies did not scale as well with larger grid networks, highlighting that more lights is only beneficial if they are coordinated well. We also saw that the main factors in congestion were dependent on the human which would be out of our control in actual practice. We saw that $v_{\max} = 5$ was near optimal whilst after a certain point it was not beneficial to increase the limit.

7.2 Limitations and Future Work

The study has several limitations that we should consider. Firstly, the cellular automaton model greatly simplifies the dynamics of real traffic into discrete cells, velocities and time steps. Additionally, the model represents driver behaviour only through stochastic rates rather than per-driver characteristics and captures vehicle heterogeneity only as a two class split of the vehicles. Although Green wave strategy performed best in our simulations, it may not always be the optimal strategy in real-world scenarios and is hard to implement in practice due to the need for precise timing.

In the future we can explore more complicated traffic optimisation techniques such as training a machine learning model to enable traffic light control, which uses real-time data to adjust traffic light timings based on current traffic conditions. We can also extend the lattice so that it supports two way roads, which would make it more realistic and allow us to explore the effects of traffic light strategies on different types of roads. We can also include varying speed limits in the system and more complicated movements with two way roads.

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